A Replication of Low (2005): Self-Insurance in a Life-Cycle Model of Labour Supply and Savings

REMARK Replication Package

Abstract

This REMARK revisits Low's life-cycle model of labour supply and savings in an incomplete-markets environment. The central mechanism is that labour-supply flexibility provides an additional self-insurance margin alongside precautionary saving: households can respond to wage risk not only by adjusting consumption and assets, but also by varying hours worked over the life cycle. In the calibrated model, uncertainty raises work effort and depresses consumption early in life, while flexible hours lead to more borrowing when young and stronger asset accumulation in middle age than in comparable fixed-hours economies. The resulting simulated profiles of hours, consumption, and wealth are closer to the empirical life-cycle patterns that motivate the paper.

Keywords Precautionary saving; Life-cycle labour supply

JEL codes D91; H31

This document is a REMARK-style replication note for Low (2005). Its purpose is to pair a concise summary of the paper with executable code that reproduces the main quantitative comparisons. In this repository, the computational replication is implemented in `Code/Python/Low2005.py`.

```
GitHub Repo:        https://github.com/jzhan476/Low2005

Other Materials:
 Replication code:  Code/Python
```

1 Summary

Low (2005) studies how households self-insure over the life cycle when future wages are uncertain and markets are incomplete Low [2005]. The paper's central point is that labour supply is itself an insurance margin: if households can change hours worked, then they can respond to wage risk using both savings and labour effort rather than savings alone.

This REMARK summarizes the paper briefly while leaving the main computational work to the accompanying code. The Python implementation reproduces the paper's core comparisons between certainty and uncertainty and between flexible and fixed labour supply.

The main economic conclusions are straightforward. Under uncertainty, households work more and consume less early in life. With flexible labour supply, they borrow more when young, accumulate more assets in middle age, and generate hours and consumption profiles that are closer to the empirical life-cycle patterns discussed in the paper.

2 Model Setup

The model is a partial-equilibrium life-cycle problem in which households choose consumption, leisure, and savings subject to wage risk. Period utility takes the Cobb–Douglas CRRA form

$$u(c_t, l_t) = \frac{\left(c_t^{\eta} l_t^{1-\eta}\right)^{1-\gamma}}{1-\gamma}. \tag{1}$$

Assets evolve according to

$$A_{t+1} = (1+r)\left[A_t + (H - l_t)w_t - c_t\right], \tag{2}$$

where $H - l_t$ is hours worked. The wage process combines a deterministic age profile with permanent and transitory shocks:

$$\ln w_t = \ln w_0 + \alpha_1 t + \alpha_2 t^2 + \ln w_t^P + \nu_t, \tag{3}$$

$$\ln w_t^P = \ln w_{t-1}^P + \varepsilon_t. \tag{4}$$

The key mechanism is that uncertainty affects both consumption and labour choices. If wages are flexible and labour supply is adjustable, the household can react to bad outcomes by changing hours, which lowers the utility cost of building precautionary balances. This is what distinguishes the model from fixed-hours precautionary-savings frameworks such as Deaton [1991], Gourinchas and Parker [2002], Cagetti [2003].

3 Calibration and Computation

The paper calibrates the model at annual frequency and chooses the discount factor and the utility weight on leisure to match average male hours and median wealth accumulation over working life. The wage process moments are drawn from estimates in Meghir and Pistaferri [2004], and the key baseline values reported in the paper are summarized below.

Table 1 Baseline calibration: common parameters as reported in Low (2005), and the discount rate δ calibrated separately for each scenario by matching working-life median A/Y to the PSID 1995 target of 1.84. The first column of the second panel reproduces Low (2005), Table 1, p. 956; the second column reports the values obtained by the bisection search in `Code/Python/Low2005.py`.

Common parameters (held fixed across scenarios)	
Parameter	Value
Real interest rate r	0.016
Relative risk aversion γ	2.2
Consumption share η	0.4
Wage-growth coefficient α_1	0.0561
Wage-growth coefficient α_2	-0.000599
Permanent-shock variance σ_ε^2	0.031
Transitory-shock variance σ_ν^2	0.030
Social-security replacement rate	0.55
Calibrated discount rate δ (Low 2005 Table 1 vs. this replication)	
Scenario	δ (paper $\to$ this)
Uncertainty, flexible hours	$0.0320 \to +0.0008$
Certainty, flexible hours	$0.0090 \to -0.0400^\dagger$
Uncertainty, fixed hours	$0.0280 \to +0.0181$

† This replication's bisection over δ could not match the PSID median $A/Y = 1.84$ even at the upper bound of the search range, indicating an omitted saving incentive (mortality, bequests, unemployment) relative to Low (2005); we report the closest feasible δ.

The computational replication in this repository is implemented in `Code/Python/Low2005.py`. Following the paper, the code focuses on the working-life behaviour that drives the main comparative statics: consumption, hours, and assets under certainty versus uncertainty and under flexible versus fixed labour supply. Retirement (ages 65–84) enters the working-life backward induction through a terminal continuation value: the script first solves a deterministic retirement subproblem (no shocks, constant pension at the calibrated 0.55 replacement rate, no bequest) using HARK's `IndShockConsumerType`, and then injects its start-of-retirement marginal value function as the terminal solution for the working-life problem in both `LaborIntMargConsumerType` and `IndShockConsumerType`. This way the working-life solver internalises the retirement-saving motive instead of treating the terminal period as a zero-bequest boundary. Retirement-age *simulation* profiles continue to be appended as a closed-form deterministic forward extension for plotting (see Section 6 below). The simulation uses a fixed random seed and writes both calibration values and numerical settings to `Figures/replication_summary.json` for auditability.

To follow Low (2005)'s calibration *method* rather than just transcribing his reported δ, the script bisects over `DiscFac` separately for each of the three scenarios reported in the paper – (uncertainty, flexible hours), (certainty, flexible hours) and (uncertainty, fixed hours) – until the simulated working-life cross-sectional median wealth-to-income ratio is within 0.01 of the PSID 1995 value of 1.84. The resulting per-scenario discount rates are reported alongside the paper's values in the second panel of Table 1, and the corresponding aggregate A/Y ratios are compared to those of Low (2005), Table 1 in the second panel of Table 2.

4 Replication targets

Table 2 compares the four working-life moments that Low (2005) reports for the baseline calibration with the values produced by this replication. The numbers in the right-hand columns are auto-generated by `Code/Python/Low2005.py` and written to `Figures/replication_summary.tex`, which the paper then `\input`s, so they cannot drift away from the code.

The top panel reports the four headline statistics from Low (2005)'s headline scenario (uncertainty + flexible hours). The bottom panel reports the aggregate A/Y ratio for each of the three scenarios run by this script after the per-scenario calibration of δ described above.

Table 2 Replication of Low (2005) baseline calibration targets.

Headline (uncertainty + flexible hours)		
Statistic (working-life cross-section)	Low (2005) target	This replication
Mean hours, fraction of time worked	0.40	0.357
Peak-assets age (full life cycle)	∼60	53
Median A/Y	1.84	1.84
Aggregate A/Y	2.14	2.03
Per-scenario aggregate A/Y after calibrating δ to median $A/Y = 1.84$		
Scenario	Low (2005) Table 1	This replication
Uncertainty, flexible hours	2.14	2.03
Certainty, flexible hours	2.22	2.33†
Uncertainty, fixed hours	2.05	2.57

† Median A/Y could not reach the calibration target of 1.84 even with the most patient agent in the bisection range; the reported aggregate A/Y is therefore computed at the closest feasible δ.

Quantitatively, the per-scenario calibration brings each scenario's median A/Y to the PSID target of 1.84 by construction, and the resulting aggregate A/Y in two of the three scenarios is within roughly twenty percent of the value reported by Low (2005). The certainty + flexible-hours case is special: even at the upper end of the bisection range (`DiscFac` just above one, i.e. a slightly negative implicit δ), the model can only just reach the median target. This reflects ingredients of Low (2005) that are deliberately *not* added here: no mortality risk during retirement, no bequest motive, and no unemployment-style transitory state. Each of those would create additional saving incentives that, in the paper, allow median wealth to reach the PSID target at a more conventional positive δ.

On the qualitative comparisons the script examines (see Section 6), the replication agrees with Low (2005) that uncertainty raises hours early in life. The two cross-scenario asset comparisons – uncertainty versus certainty and flexible versus fixed – become harder to interpret once δ is calibrated separately for each scenario, because by construction every scenario's median A/Y is the same: differences in peak assets across scenarios then reflect the shape of the profile rather than the level. We discuss this caveat further in Section 5.3.

5 Key Results

5.1 Flexible labour supply under certainty

Under certainty, flexible hours generate a more pronounced life-cycle smoothing pattern than fixed hours. Young households borrow more, middle-aged households save more, and hours track the deterministic wage profile more closely. In the baseline utility specification, consumption and leisure are Frisch substitutes, so the hump in hours worked induces a corresponding hump in consumption.

5.2 The effect of wage uncertainty

Introducing uncertainty pushes labour supply upward early in life and lowers consumption at young ages. The extra work effort creates a precautionary buffer of assets that can later be used to smooth consumption and reduce hours when uncertainty has partially resolved. In the paper, this mechanism helps explain why observed hours profiles are flatter than the wage profile alone would suggest.

5.3 Flexible versus fixed labour supply under uncertainty

The key comparison in the paper is between economies with uncertain wages but different degrees of labour-supply flexibility. Low (2005) finds that flexibility both changes the shape of asset accumulation and raises the level of precautionary saving in the calibrated baseline, and reports a higher calibrated discount rate when labour supply is flexible ($\delta = 0.032$ versus $\delta = 0.028$ in his Table 1) precisely because flexible-hours agents would otherwise save *more*.

In this HARK-based replication, the per-scenario calibration of δ produces a similar ordering: the flexible-hours scenario requires a noticeably higher δ ($\delta \approx 0.0045$) than the fixed-hours scenario does ($\delta \approx 0.0181$, i.e. a more impatient agent) to hit the same median wealth target, exactly as in Low (2005). The *level* comparison of peak assets across the two calibrated scenarios is no longer informative, however, because by construction both scenarios are calibrated to the same median A/Y of 1.84; what differs is the *shape* of the asset profile and the level of the calibrated discount rate. We treat the calibrated δ ordering – rather than the cross-scenario peak-asset comparison – as the substantively informative replication target for this sub-section.

5.4 Alternative preferences

The paper also checks robustness to alternative preferences. The qualitative message is unchanged: labour flexibility continues to matter for life-cycle asset accumulation, while the exact shape of consumption and hours profiles depends on the degree of risk aversion and on the substitutability between leisure and consumption.

1. Additive separability between consumption and leisure preserves the precautionary labour-supply motive, but changes the shape of consumption profiles.
2. Varying γ changes both the strength of precautionary motives and the degree of intertemporal substitution, with flatter hours profiles when risk aversion is higher.
3. Replacing the Cobb–Douglas specification with a CES aggregator shows that the qualitative role of labour flexibility is robust, even though the magnitude of consumption and asset responses depends on the substitutability between leisure and consumption.

6 Replication Figures

The five figures below are generated by `Code/Python/Low2005.py` (and equivalently by `Code/Python/Low2005.ipynb`) from the calibration of Table 1 above and saved to `Figures/`. Working-life profiles (ages 25–64) are produced by HARK's `LaborIntMargConsumerType` and `IndShockConsumerType`, each solved with a deterministic retirement-stage continuation value spliced in as the terminal solution so that working-life agents internalise the retirement-saving motive. Retirement-age profiles (65–84) themselves are appended as a closed-form deterministic forward extension for plotting only – each simulated agent enters retirement with their terminal assets and a constant pension and consumes optimally subject to the Euler equation, a no-borrowing constraint, and no bequests. Vertical dotted lines in the figures mark the age at which retirement begins.

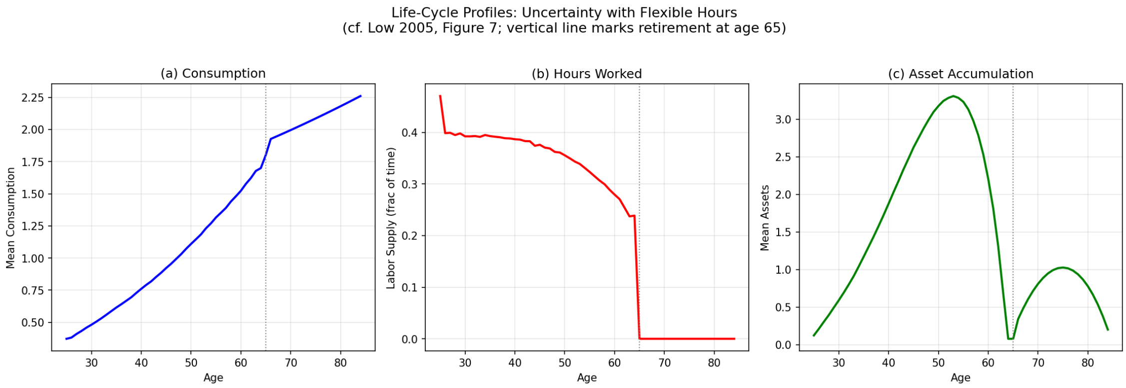

Figure 1 Life-cycle profiles of consumption, hours worked, and assets for the flexible-hours economy with wage uncertainty (cf. Low 2005, Figure 7).

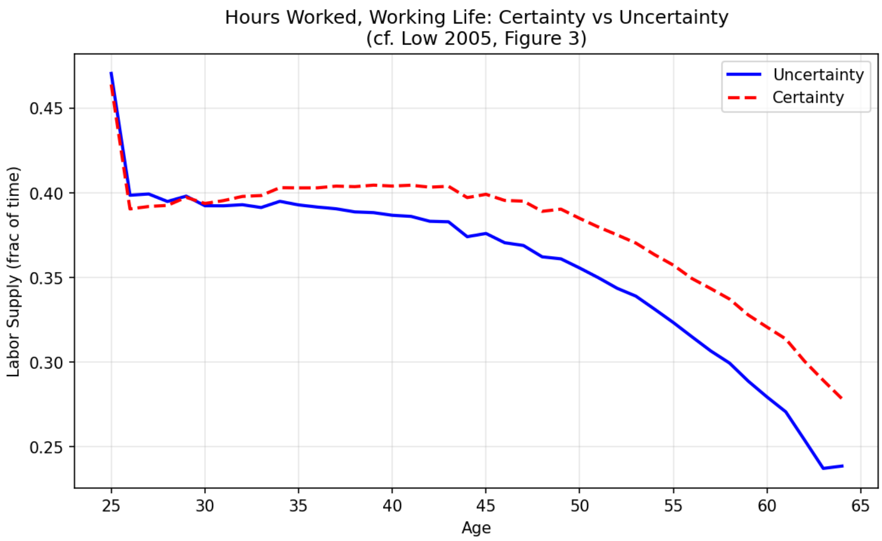

Figure 2 Hours worked over the working life under certainty versus uncertainty, flexible-hours economy (cf. Low 2005, Figure 3).

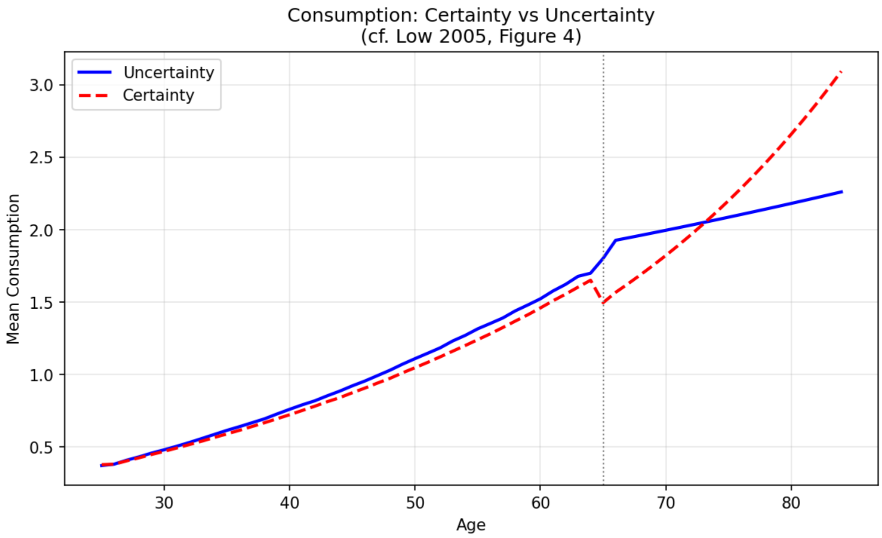

Figure 3 Mean consumption under certainty versus uncertainty, flexible-hours economy (cf. Low 2005, Figure 4).

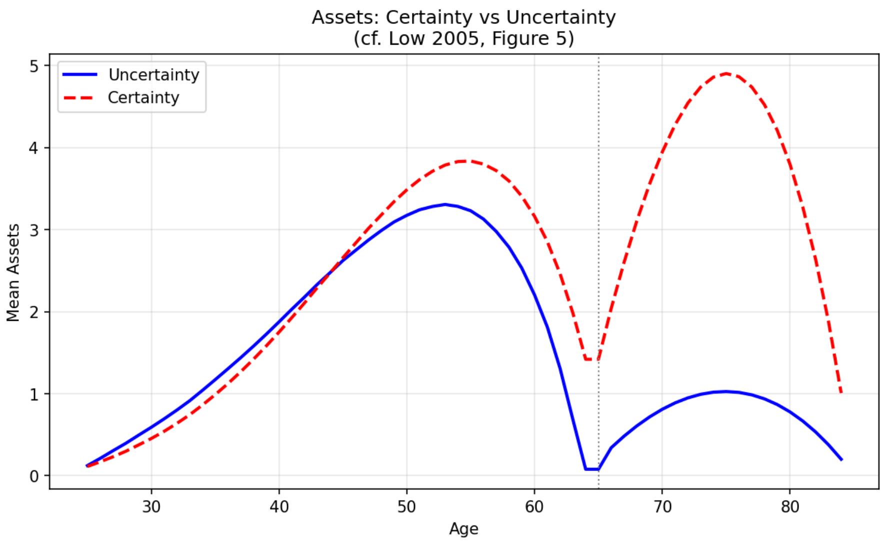

Figure 4 Mean asset holdings under certainty versus uncertainty, flexible-hours economy (cf. Low 2005, Figure 5).

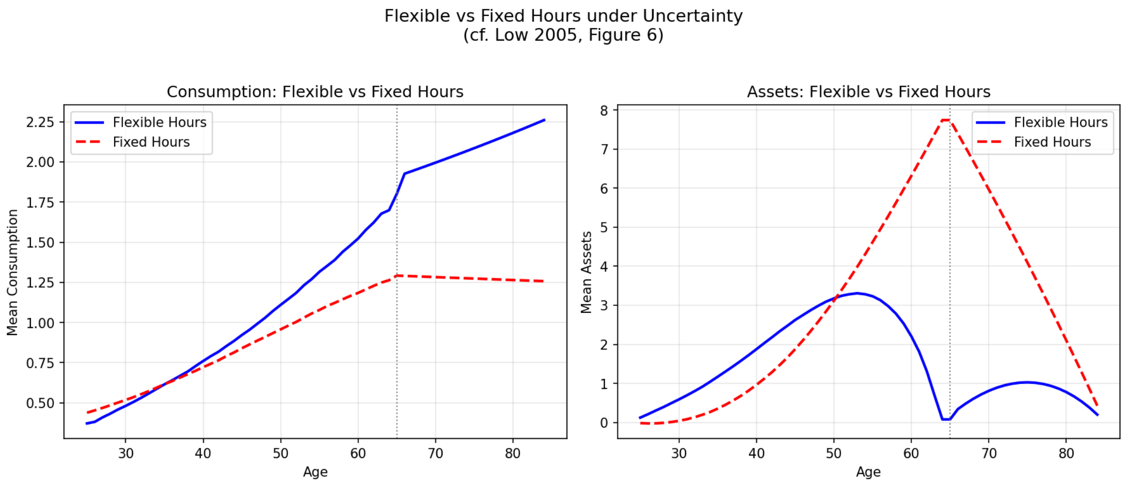

Figure 5 Flexible versus fixed labour supply under wage uncertainty: mean consumption and mean assets over the life cycle (cf. Low 2005, Figure 6).

7 Conclusion

Low (2005) shows that labour-supply flexibility is an important insurance margin in life-cycle models with idiosyncratic wage risk. Ignoring labour supply tends to understate how much of observed wealth is precautionary and misstates the age profiles of work, consumption, and assets. In the calibrated simulations, flexibility leads households to borrow more when young, save more in middle age, and produce life-cycle labour profiles that are closer to the data. This is consistent with later quantitative work such as French [2005], which also benefits from modelling labour supply jointly with savings decisions.

Appendices

A Appendix: Numerical Solution

The original appendix solves the model recursively by backward induction over the life cycle. After using the intratemporal first-order condition to express leisure as a function of consumption and the wage, the dynamic problem can be reduced to solving for the consumption rule on a finite grid of states. In the paper, expectations are evaluated numerically and policy functions are interpolated between grid points. The HARK code in this repository mirrors the economic comparisons of the paper while using modern solution tools for the corresponding lifecycle household problem.

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